

SUMMARY

AI/ML Engineer building production-oriented machine learning, agentic AI, and RAG systems with Python, FastAPI, LangGraph, PyTorch, and Docker. Delivers tested APIs, model serving, monitoring, CI/CD, guardrails, and maintainable workflows.

EXPERIENCE

Junior AI Specialist

Jun 2026 - Present

ADC Nederland | Eindhoven, The Netherlands

- Developed four production-oriented AI and data systems using Python, SQL, and APIs.
- Built LangGraph agents and 67 typed MCP tools with role-gated access.
- Implemented tenant authentication, PII controls, human confirmation, and privacy regression tests.
- Shipped FastAPI, Streamlit, Docker Compose, evaluation, and automated testing workflows.

Teaching Assistant

Sep 2021 - Jul 2025

Eindhoven University of Technology | Eindhoven, The Netherlands

- Supported 80+ students weekly in Python, machine learning, SQL, and debugging.
- Diagnosed implementation errors and coached teams toward maintainable technical solutions.

Software Technical Lead

Sep 2021 - Jul 2023

Student Team SimEnergy | Eindhoven University of Technology

- Directed six-person team integrating sensor ingestion, machine learning, and dashboards.
- Built LSTM and denoising-autoencoder models for ten appliance classes.
- Coordinated software, hardware, and business requirements and presented working prototype.

PROJECTS

Agentic Wihabo Database Chatbot with Role-Based Access

- Developed LangGraph routing for deterministic, tool-based, planned, conversational, and refusal paths.
- Integrated role-specific permissions and account-scoped credentials across operational tools.
- Reduced unnecessary LLM calls using deterministic logic and configurable model routing.
- Added persistent checkpoints, evaluation runners, guardrails, and automated tests.

Fraud Detection MLOps Pipeline - IEEE-CIS

- Built end-to-end ML pipeline for 600,000 transactions and 400+ features.
- Trained XGBoost using temporal validation, feature selection, and threshold optimization.
- Served registered models through FastAPI with MLflow tracking and drift monitoring.
- Automated testing, container builds, and CI/CD using Docker and GitHub Actions.

Offline RAG System for Legal Contracts

- Built offline RAG over 400+ legal documents on local infrastructure.
- Combined dense retrieval, BM25, reciprocal-rank fusion, and cross-encoder reranking.
- Constrained Ollama generation to retrieved context and displayed supporting sources.

EDUCATION

MSc Data Science and Artificial Intelligence | Eindhoven University of Technology

2023 - 2025

- Thesis: Reward-Free Safe Reinforcement Learning Exploration Using Different Entropy Measures

BSc Data Science | Eindhoven University of Technology and Tilburg University

2020 - 2023

- Bachelor End Project: Health Platform Text Classification Using Active Learning

Honors Academy - AI Track | Eindhoven University of Technology

2021 - 2023

- Projects: energy disaggregation, time-series modeling, visualization, and GANs

SKILLS

Programming & ML: Python, SQL, PyTorch, TensorFlow, scikit-learn, XGBoost

LLM / GenAI: LangGraph, LangChain, LangSmith, RAG, Qdrant, Ollama, MCP

Retrieval & Agents: embeddings, BM25, reranking, tool calling, model routing, guardrails

MLOps: MLflow, model registry, serving, monitoring, drift detection, CI/CD

APIs & Deployment: FastAPI, Pydantic, Docker Compose, GitHub Actions, pytest, Git

LANGUAGES

English (Fluent), Cantonese (Native), Mandarin (Fluent), Dutch (A2)